

YEAR 7 MATHEMATICS

Chapter 5A

Algebra

WORKSHEETS

5A

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

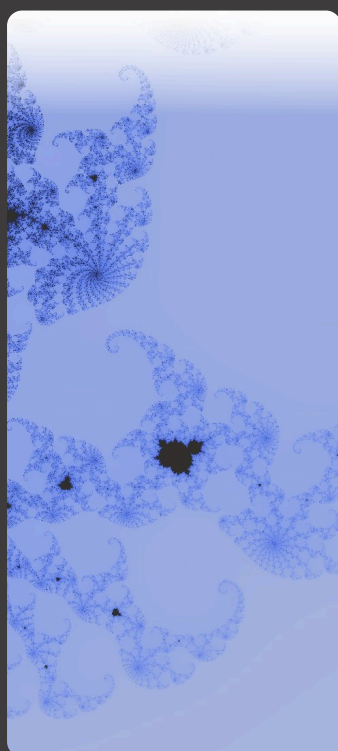
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
5.01	The commutative and associative laws	3		☐
5.02	The distributive law	4		☐
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In this chapter you will

- use the commutative, associative and distributive laws
- use variables and algebraic notation
- write expressions from words
- substitute into expressions and formulas
- add, subtract, multiply and divide algebraic terms



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is Elephant Valley, the row of trunks along the right-hand edge of the main body. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

5.01 The commutative and associative laws

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Easy

Questions 1–25

Multiply the friendly pair first (they make 10 or 100), then the other number.



WE DO

With your teacher: copy the working.

E1 $4 \times 12 \times 25$ _____

E2 $20 \times 6 \times 5$ _____

YOU DO

Your turn.

1 $3 \times 50 \times 2$ _____

300

2 $25 \times 4 \times 4$ _____

3 $13 \times 20 \times 5$ _____

4 $5 \times 28 \times 2$ _____

5 $25 \times 27 \times 4$ _____

6 $10 \times 2 \times 50$ _____

7 $14 \times 5 \times 2$ _____

8 $19 \times 50 \times 2$ _____

9 $16 \times 20 \times 5$ _____

10 $20 \times 10 \times 5$ _____

11 $13 \times 25 \times 4$ _____

12 $5 \times 4 \times 25$ _____

Add the friendly pair first (they make a multiple of 10).



WE DO

With your teacher: copy the working.

E3 $38 + 89 + 42$ _____

YOU DO

Your turn.

13 $46 + 64 + 24$ _____

134

14 $41 + 41 + 49$ _____

15 $38 + 47 + 42$ _____

16 $71 + 34 + 49$ _____

17 $88 + 49 + 42$ _____

18 $36 + 76 + 4$ _____

19 $81 + 33 + 29$ _____

20 $37 + 26 + 3$ _____

21 $23 + 41 + 47$ _____

22 $11 + 40 + 39$ _____

23 $46 + 27 + 24$ _____

24 $65 + 49 + 5$ _____

25 $51 + 53 + 49$ _____

Medium

Questions 26–41

Rearrange so the friendly numbers are together, then evaluate.



WE DO

With your teacher: copy the working.

E1 $5 \times 15 \times 2$ _____

E2 $20 \times 4 \times 5$ _____

YOU DO

Your turn.

26 $20 \times 6 \times 5$ _____

$= 20 \times 5 \times 6$

$= 100 \times 6$

$= 600$

27 $5 \times 16 \times 2$ _____

$= 5 \times 2 \times 16$

$= 10 \times 16$

$= \underline{\hspace{2cm}}$

28 $5 \times 8 \times 2$ _____

$= \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$

$= \underline{\hspace{2cm}}$

29 $5 \times 5 \times 2$ _____

30 $125 \times 15 \times 8$ _____

31 $25 \times 7 \times 4$ _____

32 $50 \times 14 \times 2$ _____

33 $125 \times 4 \times 8$ _____

34 $125 \times 18 \times 8$ _____

35 $125 \times 8 \times 8$ _____

36 $20 \times 12 \times 5$ _____

37 $25 \times 11 \times 4$ _____

38 $25 \times 15 \times 4$ _____

39 $5 \times 13 \times 2$ _____

40 $125 \times 3 \times 8$ _____

41 $50 \times 15 \times 2$ _____

Challenging

Questions 42–48

Use the commutative and associative laws to evaluate in your head. Show the order you used.



WE DO

With your teacher: copy the working.

E1 $50 \times 9 \times 3 \times 2$ _____

E2 $25 \times 11 \times 9 \times 4$ _____

YOU DO

Your turn.

42 $20 \times 5 \times 9 \times 5$ _____

$= (20 \times 5) \times (5 \times 9)$

$= 100 \times 45$

$= \underline{\hspace{2cm}}$

43 $20 \times 13 \times 2 \times 5$ _____

$= (\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}) \times (\underline{\hspace{1cm}} \times \underline{\hspace{1cm}})$

$= \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$

$= \underline{\hspace{2cm}}$

44 $5 \times 12 \times 8 \times 2$ _____

45 $5 \times 16 \times 5 \times 2$ _____

46 $50 \times 15 \times 4 \times 2$ _____

47 $5 \times 7 \times 5 \times 2$ _____

48 $50 \times 15 \times 3 \times 2$ _____

Easy

Questions 1–29

Split the bigger number into tens and ones, then multiply each part.



WE DO

With your teacher: copy the working.

E1 6×26 _____

E2 5×14 _____

YOU DO

Your turn.

1 6×29 _____

174

2 5×48 _____

3 4×41 _____

4 6×37 _____

5 6×16 _____

6 4×36 _____

7 5×47 _____

8 9×16 _____

9 4×21 _____

10 7×18 _____

11 7×33 _____

12 5×43 _____

13 8×32 _____

14 5×35 _____

15 9×14 _____

16 3×27 _____

17 5×16 _____

18 9×47 _____

19 8×22 _____

20 7×31 _____

21 6×35 _____

22 7×37 _____

23 7×38 _____

24 6×15 _____

25 4×49 _____

26 6×19 _____

27 3×24 _____

28 9×25 _____

29 6×36 _____

Medium

Questions 30–36



Use a near ten: $9 \times 34 = 10 \times 34 - 1 \times 34$.

WE DO

With your teacher: copy the working.

E1 21×40 _____

E2 99×27 _____

YOU DO

Your turn.

30 29×18 _____

$= 30 \times 18 - 1 \times 18$

$= 540 - 18$

$= 522$

31 11×23 _____

$= 10 \times 23 + 1 \times 23$

$= 230 + 23$

$=$ _____

32 29×29 _____

$=$ _____ $\times$ _____ $-$ _____ $\times$ _____

$=$ _____ $-$ _____

$=$ _____

33 21×14 _____

34 9×23 _____

35 19×42 _____

36 11×41 _____

Challenging

Questions 37–46

Expand the brackets: multiply each term inside by the number outside.



WE DO

With your teacher: copy the working.

E1 $7(y - 5)$ _____

E2 $5(t + 9)$ _____

YOU DO

Your turn.

37 $6(t - 5)$ _____

$= 6 \times t - 6 \times 5$

$=$ _____ $t -$ _____

38 $6(y - 5)$ _____

$=$ _____ $\times y -$ _____ $\times$ _____

$=$ _____ $y -$ _____

39 $3(n + 1)$ _____

40 $9(n + 1)$ _____

41 $5(m - 1)$ _____

42 $9(m + 1)$ _____

43 $3(y + 2)$ _____

44 $6(m + 3)$ _____

45 $3(c - 1)$ _____

46 $7(y - 12)$ _____

Easy

Questions 1–29

Simplify: write the number first, then the letters in alphabetical order.

WE DO With your teacher: copy the working.

E1 $p \times p$

E2 $b \times b$

YOU DO Your turn.

1 $y \times x \times 3$

$3xy$

2 $n \times c \times 3$

3 $x \times b \times 8$

4 $p \times b \times 5$

5 $y \times 12$

6 $n \times 11$

7 $y \times a \times 11$

8 $x \times 11$

9 $c \times c$

10 $c \times 12$

11 $a \times a$

12 $t \times t$

13 $x \times t \times 6$

14 $x \times 3$

15 $t \times 12$

16 $n \times 6$

17 $m \times m$

18 $y \times x \times 10$

19 $n \times n$

20 $k \times 3$

21 $m \times 2$

22 $p \times 10$

23 $k \times 6$

24 $k \times 7$

25 $y \times 7$

26 $x \times m \times 6$

27 $n \times 5$

28 $y \times p \times 8$

29 $m \times 6$

Medium

Questions 30–39

Write in expanded form (with $\times$ signs), then simplify another way round.

WE DO With your teacher: copy the working.

E1 $2bp$

E2 $7ap$

YOU DO Your turn.

30 $8bc$

$= 8 \times b \times c$

$= 8cb$ (also correct)

31 $3kp$

$= 3 \times k \times p$

$= ______ pk$ (also correct)

32 $2ck$

$= ______ \times c \times k$

$= ______ kc$ (also correct)

33 $6by$

34 $8bp$

35 $7kt$

36 $6ck$

37 $9bk$

38 $3at$

39 $2mx$

Challenging

Questions 40–49

Simplify. Use index notation for repeated letters.

WE DO With your teacher: copy the working.

E1 $7 \times t \times y \times t \times 3$

E2 $4 \times a \times c \times a \times 4$

YOU DO Your turn.

40 $9 \times b \times n \times b \times 5$

$= 9 \times 5 \times b \times b \times n$

$= ______ b^2n$

41 $8 \times k \times n \times k \times 2$

$= ______ \times ______ \times k \times k \times n$

$= ______ k^2n$

42 $5 \times b \times y \times b \times 4$

43 $9 \times m \times x \times m \times 4$

44 $3 \times t \times y \times t \times 5$

45 $8 \times m \times p \times m \times 4$

46 $3 \times n \times p \times n \times 5$

47 $8 \times c \times x \times c \times 3$

48 $5 \times m \times p \times m \times 4$

49 $2 \times t \times x \times t \times 2$

Easy

Questions 1–30

Write an expression for each description.



WE DO With your teacher: copy the working.

E1 4 more than a _____

E2 7 less than a _____

YOU DO Your turn.

1 11 times p 11p

2 c divided by 10 _____

3 the sum of c and 10 _____

4 double x _____

5 b squared _____

6 c decreased by 5 _____

7 5 more than x _____

8 11 less than x _____

9 4 times n _____

10 k divided by 7 _____

11 the sum of k and 8 _____

12 double c _____

13 c squared _____

14 c decreased by 12 _____

15 6 more than a _____

16 2 less than k _____

17 9 times t _____

18 n divided by 5 _____

19 the sum of a and 9 _____

20 double b _____

21 k squared _____

22 b decreased by 8 _____

23 7 more than n _____

24 2 less than c _____

25 6 times b _____

26 m divided by 12 _____

27 the sum of k and 5 _____

28 double k _____

29 a squared _____

30 m decreased by 4 _____

Medium

Questions 31–40

Write an expression. Do the first operation, then the second.



WE DO With your teacher: copy the working.

E1 3 less than twice k _____

E2 11 more than 3 times c _____

YOU DO Your turn.

31 6 times b , take away 11

first: $6b$

$= 6b - 11$

32 x squared, plus 9

first: x^2

$= x^2 + \underline{\hspace{1cm}}$

33 7 less than the product of p and y

first: _____

$= py - \underline{\hspace{1cm}}$

34 8 less than twice a _____

35 6 more than 7 times c _____

36 5 times k , take away 2 _____

37 a squared, plus 11 _____

38 9 less than the product of a and p

$\underline{\hspace{1cm}}$

39 4 less than twice a _____

40 11 more than 5 times c _____

Challenging

Questions 41–47

Write an expression for each amount. Say what the letter stands for.



WE DO With your teacher: copy the working.

E1 the cost of m pens at \$15 each, plus \$18 for a folder

E2 the number of legs on x spiders and 13 birds

E3 \$100 shared equally by n people

YOU DO Your turn.

41 the cost of y pens at \$15 each, plus \$18 for a folder

y = number of pens

cost = $\underline{\hspace{1cm}}y + \underline{\hspace{1cm}}$ dollars

42 the number of legs on t spiders and 14 birds

$\underline{\hspace{1cm}}$

legs = $\underline{\hspace{1cm}}t + \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}t + \underline{\hspace{1cm}}$

43 \$70 shared equally by p people

44 the cost of b pens at \$8 each, plus \$11 for a folder

45 the number of legs on n spiders and 4 birds

46 \$130 shared equally by x people

47 the cost of t pens at \$9 each, plus \$12 for a folder

Easy

Questions 1–30

Substitute the value, then evaluate.



WE DO With your teacher: copy the working.

E1 $2n, n = 8$

E2 $2a, a = 9$

YOU DO Your turn.

1 $x + 3, x = 5$

8

2 $x + 8, x = 8$

3 $4p, p = 8$

4 $4n + 1, n = 10$

5 $4n, n = 7$

6 $2p, p = 5$

7 $n + 9, n = 9$

8 $8p, p = 6$

9 $7a + 1, a = 5$

10 $7a + 1, a = 12$

11 $6n, n = 4$

12 $9n + 1, n = 3$

13 $x + 8, x = 12$

14 $4a + 1, a = 10$

15 $5n + 1, n = 4$

16 $7x + 1, x = 10$

17 $2p + 1, p = 8$

18 $a + 3, a = 12$

19 $8p, p = 8$

20 $p + 4, p = 4$

21 $5x, x = 11$

22 $8p + 1, p = 11$

23 $4x + 1, x = 12$

24 $8x, x = 2$

25 $a + 2, a = 6$

26 $3x + 1, x = 2$

27 $5a, a = 5$

28 $n + 4, n = 11$

29 $p + 6, p = 2$

30 $n + 9, n = 4$

Medium

Questions 31–40

Substitute the values given. Write the numbers in first, then evaluate.

WE DO With your teacher: copy the working.

E1 $ab, a = 9, b = 5$

E2 $2a + b, a = 8, b = 3$

YOU DO Your turn.

31 $3b - a, a = 7, b = 8$

$= 3 \times 8 - 7$

$= 17$

32 $a + 2b, a = 7, b = 9$

$= 7 + 2 \times 9$

$= \underline{\hspace{2cm}}$

33 $5a - b, a = 6, b = 8$

$= \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}$

34 $b^2, a = 3, b = 7$

35 $a^2 + b, a = 6, b = 4$

36 $4(a + b), a = 3, b = 7$

37 $ab, a = 9, b = 9$

38 $2a + b, a = 2, b = 9$

39 $3b - a, a = 6, b = 4$

40 $a + 2b, a = 5, b = 8$

Challenging

Questions 41–45

Substitute into the formula. Write the formula, then the numbers, then the answer.



WE DO With your teacher: copy the working.

E1 Find A when $A = lw, l = 11, w = 4$.

E2 Find P when $P = 2l + 2w, l = 3, w = 2$.

E3 A plumber charges $C = 30h + 19$ dollars. Find C when $h = 3$.

YOU DO Your turn.

41 Find A when $A = lw, l = 14, w = 10$.

$A = lw$

$A = 14 \times 10$

$A = \underline{\hspace{2cm}}$

42 Find P when $P = 2l + 2w, l = 11, w = 3$.

$P = \underline{\hspace{2cm}}l + \underline{\hspace{2cm}}w$

$P = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} + \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

$P = \underline{\hspace{2cm}}$

43 A plumber charges $C = 30h + 19$ dollars. Find C when $h = 4$.

44 Find A when $A = lw, l = 12, w = 8$.

45 Find P when $P = 2l + 2w, l = 8, w = 12$.

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9a

29 $11c - 6c$

39 $7c + 7n + 5c + 2n$

49 $6am + 3a^2 - 9am + 3a^2 + 9$

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30 $8k \times 8p$

52 $3bm \times 6b$

59 $-4b \times 2n \times 3b$

Easy

Questions 1–30

Divide the numbers. Keep the letter.



WE DO With your teacher: copy the working.

E1 $18x \div 2$ _____

E2 $24a \div 3$ _____

YOU DO Your turn.

1 $12b \div 3$ 4b

2 $35k \div 5$ _____

3 $18b \div 9$ _____

4 $48m \div 6$ _____

5 $42n \div 7$ _____

6 $20b \div 4$ _____

7 $63a \div 7$ _____

8 $64c \div 8$ _____

9 $48t \div 6$ _____

10 $16b \div 4$ _____

11 $27k \div 3$ _____

12 $40k \div 5$ _____

13 $12k \div 6$ _____

14 $6b \div 2$ _____

15 $15n \div 3$ _____

16 $72m \div 9$ _____

17 $56b \div 7$ _____

18 $14m \div 2$ _____

19 $20x \div 4$ _____

20 $27t \div 9$ _____

21 $21n \div 3$ _____

22 $48x \div 6$ _____

23 $12n \div 2$ _____

24 $21p \div 3$ _____

25 $56y \div 7$ _____

26 $12t \div 4$ _____

27 $24n \div 4$ _____

28 $16p \div 8$ _____

29 $10c \div 2$ _____

30 $14a \div 7$ _____

Medium

Questions 31–50

Write as a fraction, then cancel numbers and letters.

WE DO With your teacher: copy the working.

E1 $12ny \div 4n$ _____

E2 $18m \div 2m$ _____

YOU DO Your turn.

31 $16a \div 4a$

$= \frac{16a}{4a}$

$= 4$

32 $42px \div 6x$

$= \frac{42px}{6x}$

$= 7p$

33 $15x \div 5x$

$= \frac{15x}{5x}$

$= 3$

34 $36ck \div 6c$ _____

35 $6ap \div 3p$ _____

36 $36p \div 4p$ _____

37 $48p \div 6p$ _____

38 $8py \div 2p$ _____

39 $6my \div 2m$ _____

40 $36a \div 4a$ _____

41 $36c \div 6c$ _____

42 $36ky \div 4y$ _____

43 $21bm \div 3m$ _____

44 $12ab \div 3b$ _____

45 $16p \div 2p$ _____

46 $36mp \div 4m$ _____

47 $21bk \div 3k$ _____

48 $8n \div 4n$ _____

49 $27a \div 3a$ _____

50 $18y \div 2y$ _____

Challenging

Questions 51–56

Simplify the fraction. Cancel the numbers, then the letters ($a^2 \div a = a$).



WE DO With your teacher: copy the working.

E1 $\frac{30m2y}{5my}$ _____

E2 $\frac{12a2x}{6ax}$ _____

YOU DO Your turn.

51 $\frac{54p2y}{6py}$

numbers: $54 \div 6 = 9$

letters: $p^2y \div py = p$

$= 9p$

52 $\frac{16b2x}{4bx}$

numbers: $16 \div 4 = 4$

letters: $b^2x \div bx = b$

$= 4b$

53 $\frac{45m2t}{5mt}$ _____

54 $\frac{42b2y}{6by}$ _____

55 $\frac{8k2x}{2kx}$ _____

56 $\frac{6m2x}{3mx}$ _____